7. We apply the normal equations.

$$\hat{x} = \alpha_1 y^1 + \alpha_2 y^2$$

where $\quad G^T \alpha = b \quad$ and

$$G = \begin{bmatrix} \langle y^1, y^1 \rangle & \langle y^2, y^1 \rangle \\ \langle y^1, y^2 \rangle & \langle y^2, y^2 \rangle \end{bmatrix}$$

$$b = \begin{bmatrix} \langle x, y^1 \rangle \\ \langle x, y^2 \rangle \end{bmatrix}$$

Doing the calculations, we have

$$\cancel{\langle y^1, y^2 \rangle = tr\left(\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\right) = tr\left(\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}\right) = 3}$$

$$\cancel{\langle y^1, y^2 \rangle = \langle y^2, y^1 \rangle = tr(}$$

$$\cancel{\langle y^1, y^1 \rangle = tr\left(\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}\right) = tr\big(\big[}$$

$$\langle y^1, y^1\rangle = tr\left(\begin{bmatrix}1 & 2\\ 0 & 0\end{bmatrix}\begin{bmatrix}1 & 0\\ 2 & 0\end{bmatrix}\right) = tr\left(\begin{bmatrix}5 & 0\\ 0 & 0\end{bmatrix}\right) = 5$$

$$\langle y^1, y^2\rangle = \langle y^2, y^1\rangle = tr\left(\begin{bmatrix}1 & 2\\ 0 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\right) = tr\left(\begin{bmatrix}3 & 3\\ 0 & 0\end{bmatrix}\right) = 3$$

$$\langle y^2, y^2\rangle = tr\left(\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\right) = tr\left(\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}\right) = 4$$

$$\langle x, y^1\rangle = tr\left(\begin{bmatrix}0 & 2\\ -1 & 0\end{bmatrix}\begin{bmatrix}1 & 0\\ 2 & 0\end{bmatrix}\right) = tr\left(\begin{bmatrix}4 & 0\\ -1 & 0\end{bmatrix}\right) = 4$$

$$\langle x, y^2\rangle = tr\left(\begin{bmatrix}0 & 2\\ -1 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\right) = tr\left(\begin{bmatrix}2 & 2\\ -1 & -1\end{bmatrix}\right) = 1$$

$$\therefore \quad \begin{bmatrix}5 & 3\\ 3 & 4\end{bmatrix}\begin{bmatrix}\alpha_1\\ \alpha_2\end{bmatrix} = \begin{bmatrix}4\\ 1\end{bmatrix}$$

$$\therefore \quad \begin{bmatrix}\alpha_1\\ \alpha_2\end{bmatrix} = \frac{1}{11}\begin{bmatrix}4 & -3\\ -3 & 5\end{bmatrix}\begin{bmatrix}4\\ 1\end{bmatrix} = \frac{1}{11}\begin{bmatrix}13\\ -7\end{bmatrix} = \begin{bmatrix}1.18\\ -0.64\end{bmatrix}$$

$$\hat{x} = \frac{13}{11}\begin{bmatrix}1 & 0\\ 2 & 0\end{bmatrix} - \frac{7}{11}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} = \begin{bmatrix}\frac{6}{11} & -\frac{7}{11}\\ \frac{19}{11} & -\frac{7}{11}\end{bmatrix}$$

8. Let $\gamma := d(x, M)$, and suppose that $m_1, m_2 \in M$ satisfy $\|x - m_i\| = \gamma$.

<u>To show</u> $m_1 = m_2$ when the norm is strict.

Because M is a subspace, $\frac{m_1 + m_2}{2} \in M$.

Hence,

$$\gamma = \inf_{y \in M} \|x - y\| \le \left\|x - \frac{m_1 + m_2}{2}\right\|$$

$$= \left\| \frac{x - m_1}{2} + \frac{x - m_2}{2} \right\|$$

$$\le \frac{1}{2}\|x - m_1\| + \frac{1}{2}\|x - m_2\|$$

$$= \frac{\gamma}{2} + \frac{\gamma}{2}$$

$$= \gamma.$$

Hence, $\|(x - m_1) + (x - m_2)\| = \|x - m_1\| + \|x - m_2\|$

Because the norm is strict, $\exists\, \alpha \geq 0$ such that either (i) $(x-m_1) = \alpha(x-m_2)$ or (ii) $(x-m_2) = \alpha(x-m_1)$.

In either case, we deduce from $\gamma = \|x-m_1\| = \|x-m_2\|$, that $\gamma = \alpha\gamma$, and, ~~thus~~ because $\gamma \neq 0$, we have $\alpha = 1$.

With $\alpha = 1$, we have $x - m_1 = x - m_2$, and thus $m_1 = m_2$.

$\square$

9.

(a) Not strictly normed. Let

$$x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then $\|x+y\|_1 = 2 = \|x\|_1 + \|y\|_1$,

but there does not exist any $\alpha \geq 0$ such that either $x = \alpha y$ or $y = \alpha x$.

(c) Not strictly normed. Let

$$x = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ and } y = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

Then $6 = \|x+y\|_\infty = \|x\|_\infty + \|y\|_\infty$,

but there does not exist any $\alpha \geq 0$ such that either $x = \alpha y$ or $y = \alpha x$.

(b) Strictly normed The result is true for any norm induced by an inner product. Hence we give the proof for $\|x\| = \langle x,x\rangle^{1/2}$.

Let $x, y \in X$

Case 1 Either x or y is zero. Then $\|x+y\| = \|x+y\|$ is always true and either $x = 0 \cdot y$ or $y = 0 \cdot x$ holds. $\square$

Case 2 Both $x \neq 0$ and $y \neq 0$, but $\{x, y\}$ is linearly dependent. Then ~~Suppose~~ $x = \alpha y$ for some $\alpha \in \mathbb{R}$. It follows that

$$\|x+y\| = \|1+\alpha\| \cdot \|y\|$$

and

$$\|x\| + \|y\| = (1 + |\alpha|)\,\|y\|.$$

Because $y \neq 0$, we have $\|x+y\| = \|x\| + \|y\| \iff$

$|1+\alpha| = 1 + |\alpha|$. $\iff \alpha \geq 0$.

~~Deduce that $\alpha \geq 0$. Hence,~~

Hence, $\|x+y\| = \|x\| + \|y\| \iff x = \alpha y,\ \alpha \geq 0.$

$\square$

<u>Case 3</u> $\{x, y\}$ is linearly independent.

By the Gram-Schmidt procedure, there exists $v \in X$ such that

$x \perp v$ and $\operatorname{span}\{x, y\} = \operatorname{span}\{x, v\}$.

Write $y = \alpha_1 x + \alpha_2 v$, so that

$$x + y = (1 + \alpha_1)\,x + \alpha_2 v$$

$$\|x+y\| = \|x\| + \|y\|$$ if, and only if

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2\|x\|\cdot\|y\|$$

By the Pythagorean Theorem,

$$\begin{aligned}\|x+y\|^2 &= \|(1+\alpha_1)x + \alpha_2 v\|^2 \\ &= (1+\alpha_1)^2\|x\|^2 + (\alpha_2)^2\|v\|^2 \\ &= [1 + 2\alpha_1 + (\alpha_1)^2]\|x\|^2 + (\alpha_2)^2\|v\|^2\end{aligned}$$

Furthermore

$$\|y\|^2 = (\alpha_1)^2\|x\|^2 + (\alpha_2)^2\|v\|^2$$

Hence, $\|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2\|x\|\cdot\|y\|$

if, and only if,

$$2\alpha_1\|x\|^2 = 2\|x\|\cdot\|y\|$$

Because $||x|| \neq 0$ from $\{x, y\}$ linearly independent, we have

$$\alpha_1 ||x|| = ||y||$$

$\therefore$ $\alpha_1 \geq 0$. Moreover

$$(\alpha_1)^2 ||x||^2 = ||y||^2 = (\alpha_1)^2 ||x||^2 + (\alpha_2)^2 ||u||^2$$

if, and only if, $\alpha_2 = 0$.

Hence, $||x+y|| = ||x|| + ||y||$ if, and only if $y = \alpha_1 x$, $\alpha_1 \geq 0$.

$\square$